

Susmita Sarkar, Ph.D.

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Work Authorization: Eligible to work in the United States | No sponsorship required

PROFESSIONAL SUMMARY

Researcher with 3+ years of professional experience in survey design, programme management, social and environmental research. Experienced in working with field and secondary data, developing and tracking indicators, and communicating findings through data visualizations, GIS maps, and research reports. Currently pursuing M.S. in Social Data Analytics and Research at UT Dallas. Strong background in coordinating research activities, maintaining project documentation, collaborating with diverse stakeholders, and communicating findings through reports, presentations, data visualizations, and GIS maps.

EDUCATION

University of Texas at Dallas

Richardson, TX

M.S. in Social Data Analytics and Research.

Present

Current Coursework: Data Visualization, Python Programming for Social Science, Internet Map Information Processing

Jamia Millia Islamia

New Delhi, India

Ph.D. in Geography (UGC-JRF/SRF). Dissertation: Vulnerability Mapping of Urban Hazards with a Focus on Seismic Hazards in Delhi.

2020

Jamia Millia Islamia

New Delhi, India

M.A. in Geography.

2013

University of Delhi, Kamala Nehru College

New Delhi, India

B.A. (Honours) in Geography.

2011

RESEARCH EXPERIENCE

Doctoral Researcher, Geography

2014 – 2020

Jamia Millia Islamia

New Delhi, India

- Ph.D. dissertation research on seismic social vulnerability in Delhi using satellite imagery, primary field-survey data, and secondary datasets.
- Designed and conducted field surveys to collect socioeconomic, building-structural, knowledge, perception, and emergency-preparedness data across Delhi.
- Applied statistical models to analyze survey data and derive indicators of vulnerability.
- Used GIS and remote sensing techniques to integrate analytical results and produce seismic vulnerability maps of Delhi.
- Integrated socioeconomic and spatial data to assess patterns of vulnerability to seismic hazards.

PROFESSIONAL EXPERIENCE

Senior Programme Officer

September 2024 – September 2025

Centre for People's Forestry

Hyderabad, India

- Led and oversaw projects in West Bengal, including project planning, monitoring, implementation, and reporting.
- Coordinated research activities and supported work across other state projects as required.
- Managed donor outreach and grant management, including funding pipelines, reporting requirements, and preparation of donor reports.
- Coordinated research, stakeholder consultations, knowledge-sharing workshops, and policy engagements.
- Represented the organization at academic, government, donor, multi-stakeholder, and policy forums.
- Prepared web map of project locations, supporting communication of field-level activities.

Associate Analyst

February 2022 – April 2023

Legal Initiative for Forest and Environment

Kolkata, India

- Conducted research on environmental issues and violations of environmental laws to support advocacy and policy interventions.
- Major Research Project: Assessed the functioning of the Compensatory Afforestation Fund Management and Planning Authority (CAMPA) Act and Forest Clearances in Eastern India
- Additional Research Projects: Conducted research on wetland conservation, Eco-Sensitive Zone and investigated brick kiln operations in areas of Eastern India.
- Prepared research reports and policy critiques.

Assistant Research Officer

October 2020 – February 2021

INCLIN Trust International

New Delhi, India

- Trained field workers in the collection of socioeconomic data for public health studies.
- Prepared and updated village maps for surveys and GIS-based reporting.

FREELANCE & CONSULTING EXPERIENCE

Consultant Subject Matter Expert, Social Science and Geography

January 2025 – Present

Tata Consultancy Services iON

India

- Provide subject-matter expertise in question creation and review for Social Science and Geography examinations in India.

Project Associate

March 2021 – January 2022

Dronexia Innovations Pvt. Ltd.

New Delhi, India

- Mapped and digitized remote sensing and UAV-based datasets and supported spatial analysis and reporting.

TECHNICAL SKILLS

Geospatial Tools: ArcGIS Pro, ArcGIS Online (including web mapping), ArcGIS Desktop, QGIS, Google Earth Engine, ERDAS Imagine, Global Mapper, Pix4Dmapper

Data Analysis Tools: Python, R, SPSS, STATA

AWARDS & HONORS

- Keith Lankford Taylor Scholars Scholarship and Competitive Scholarship Tuition Waiver, M.S. in Social Data Analytics and Research, University of Texas at Dallas, 2026
- University Grants Commission National Eligibility Test (Junior Research Fellowship) Research Fellowship (National-level) for doctoral studies
- Best Presentation Award (Poster category), International Conference on Remote Sensing and GIS Applications in Geosciences, Aligarh Muslim University, 2017
- Amul Vidya Bhushan Award for academic excellence at the Senior Secondary Level across India, 2008

PUBLICATIONS

- Taqi, M., Sarkar, S., & Khan, M.M.A. (2020). Analyzing the Disparities in the Coverage of Maternal and Child Health Services: A District Level Cross-Sectional Analysis of Jammu and Kashmir. *Indian Journal of Public Health*. DOI: 10.4103/ijph.IJPH_103_19
- Nashier, S., & Sarkar, S. (2020). Urban Agriculture Practices and Benefits: A Study of Unplanned Localities of Najafgarh Zone, NCT of Delhi. *Urban Environment: Issues and Challenges*. ISBN: 9382847995.
- Taqi, M., Bidhuri, S., Sarkar, S., Ahmad, W.S., & Wangchok, P. (2017). Rural Healthcare Infrastructural Disparities in India: A Critical Analysis of Availability and Accessibility. *Journal of Multidisciplinary Research in Healthcare*, Chitkara University. DOI: 10.15415/jmrh.2017.32011
- Parveen, S., & Sarkar, S. (2018). Land Use Land Cover Mapping with Change Detection: A Spatio-Temporal Analysis of NCT of Delhi from 1981 to 2015. *International Journal of Geomatics and Geosciences*.

CONFERENCE PRESENTATIONS

- Seismic Hazard and Micro-Risk Zonation in NCT of Delhi. RSU International Research Conference on Science and Technology, Social Sciences, and Humanities, Rangsit University, Thailand, 2019. Oral presentation.
- Status of Production and Consumption of Urban Agriculture: A Household Level Study of Selected Localities in West Delhi. National Conference on Urban Environment in the 21st Century: Issues and Challenges, Jamia Millia Islamia, New Delhi, 2018. Oral presentation.
- Hazard and Risk Zone Identification in a Multi-hazard Urban Environment. International Conference on Remote Sensing and GIS Applications in Geosciences, Aligarh Muslim University, 2017. Poster presentation.
- Literacy Status Amongst Female Workers in a Medium Sized City: A Case Study of Sehore Municipality, Madhya Pradesh. National Conference on Geoinformatics for Natural Resource Management, Jamia Millia Islamia, New Delhi, 2017. Oral presentation.
- Analysis of Dynamic Change Rate of Urban Landscape: A Study of Kolkata Metropolitan Area, West Bengal. International Conference on Spatial Decision Support System for UN Sustainable Development Goals, Kalindi College, University of Delhi, 2017. Oral presentation.
- A Spatio-Temporal Assessment of Urban Growth Patterns and Its Impact on the Climatic Conditions of Kolkata, West Bengal. International Conference on Land Use, Climate Extremes and Disaster Risk Reduction, Shaheed Bhagat Singh College, University of Delhi, 2016. Oral presentation.

ACADEMIC TRAINING & CERTIFICATIONS

- Training in UAV Remote Sensing (two-week), *North-Eastern Space Application Centre*, 2023
- Pre-Symposium Tutorial on Hyper-Spectral Remote Sensing and Its Applications, *Indian Society of Remote Sensing*, 2016
- Training in Remote Sensing and GIS (One-Week), *Regional Remote Sensing Centre*, ISRO, Kolkata, 2013
- Workshop on Quantitative Approaches in the Social Sciences: Survey, Experiment and Content Analysis, *Jamia Millia Islamia*, 2017
- Research Methodology Workshop on Geography and Environment with SPSS Training, *Jamia Millia Islamia*, 2017
- GIAN Workshop on Global Cities Sustainability and Management of the Urban Ecosystem, *Jamia Millia Islamia*, 2016
- Research methodology workshops at *Jamia Millia Islamia*.
- Certificate Courses in R Programming, STATA, and Python (Online)